

418-hw4

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Problem 1

```
# a)
# H0:  $\mu_1 = \mu_2 = \mu_3$  OR  $\tau_1 = \tau_2 = \tau_3 = 0$ 
# vs. H1: At least one inequality
# One-way (cooking method) MANOVA model:  $y_{ij} = \mu + \tau_i + \epsilon_{ij}$ 
# Where  $y$  is the response variable vector,  $\mu$  is the hypothetical overall mean,
#  $\tau$  is between study variability, and  $\epsilon$  is random error.
#  $j$  = observation,  $i$  = group number.  $_{ij}$  refers to  $j$ th observation in  $i$ th group

# b)
fish = read.csv('C:/Users/jacma/Downloads/Fish_Table6.17.csv')
n = 12
methods = fish[,1] # variable vectors
aroma = fish[,2]
flavor = fish[,3]
texture = fish[,4]
moisture = fish[,5]
fish_manova = manova(cbind(aroma,flavor,texture,moisture)~methods,data=fish)
summary(fish_manova)
```

```
##           Df  Pillai approx F num Df den Df    Pr(>F)
## methods      1 0.71623   19.561      4    31 4.016e-08 ***
## Residuals 34
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(fish_manova,test='Wilks')
```

```
##           Df   Wilks approx F num Df den Df    Pr(>F)
## methods      1 0.28377   19.561      4    31 4.016e-08 ***
## Residuals 34
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(fish_manova,test='Roy')
```

```
##           Df   Roy approx F num Df den Df   Pr(>F)
## methods    1 2.524   19.561     4    31 4.016e-08 ***
## Residuals 34
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(fish_manova,test='Hotelling-Lawley')
```

```
##           Df Hotelling-Lawley approx F num Df den Df   Pr(>F)
## methods    1           2.524   19.561     4    31 4.016e-08 ***
## Residuals 34
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The results of all four tests are identical: Reject H_0 and conclude that at least one of the mean vectors is significantly different to the others. The only difference between the tests is the test statistic used.

c)
The dimension of the data is 3, because there are 3 groups. This means there are three eigenvalues. However, it can be seen that the test statistic for Hotelling-Lawley and Roy's tests are the same, implying the maximum eigenvalue is equivalent to the sum of the eigenvalues. This means all other eigenvalues are 0, and this can be verified by checking the formulas for Pillai's trace and Wilk's lambda using eigenvalues 0, 0, and 2.524. Since only one eigenvalue is non-zero, the effectivedimensionality is 1.

Problem 2a

```
# a)
#  $H_0$ :  $\mu_{Sk} = \mu_{Vk}$  (mean of  $k$ th variable does not differ between factors  $S$  or  $V$ )
# vs.  $H_1$ :
beans = read.csv('C:/Users/jacma/Downloads/Snapbean_table6.18.csv')
sowdate = beans[,1]
variety = beans[,2]
early = beans[,4]
sla = beans[,5]
total = beans[,6]
avg = beans[,7]
bean_data = data.frame(sowdate,variety,early,sla,total,avg)
variable_names = list('Sowdate','SLA','Total','Average')
for (i in 3:6) {
  print(variable_names[i-2])
  sow_aov = aov(bean_data[,i]~sowdate,data=bean_data)
  print(summary(sow_aov))
  var_aov = aov(bean_data[,i]~variety,data=bean_data)
  print(summary(var_aov))
  cat('\n')
}
```

```

## [[1]]
## [1] "Sowdate"
##
##           Df Sum Sq Mean Sq F value Pr(>F)
## sowdate      1  715.5   715.5   230.5 <2e-16 ***
## Residuals    58  180.0     3.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##           Df Sum Sq Mean Sq F value Pr(>F)
## variety       1    0.1    0.081   0.005  0.943
## Residuals    58  895.4  15.438
##
## [[1]]
## [1] "SLA"
##
##           Df Sum Sq Mean Sq F value Pr(>F)
## sowdate      1  164.6  164.58   178.4 <2e-16 ***
## Residuals    58   53.5    0.92
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##           Df Sum Sq Mean Sq F value Pr(>F)
## variety       1   3.19   3.192   0.862  0.357
## Residuals    58 214.89   3.705
##
## [[1]]
## [1] "Total"
##
##           Df Sum Sq Mean Sq F value Pr(>F)
## sowdate      1  691.0   691.0   473.1 <2e-16 ***
## Residuals    58   84.7    1.5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##           Df Sum Sq Mean Sq F value Pr(>F)
## variety       1    0.1    0.081   0.006  0.938
## Residuals    58  775.6  13.373
##
## [[1]]
## [1] "Average"
##
##           Df Sum Sq Mean Sq F value Pr(>F)
## sowdate      1 31273  31273   92.93 1.19e-13 ***
## Residuals    58 19518    337
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##           Df Sum Sq Mean Sq F value Pr(>F)
## variety       1     4     4.2   0.005  0.945
## Residuals    58 50787  875.6

```

*# For each of the four variables, sowdate is extremely significant, whereas
variety is not. This suggests that sowdate is a more influential factor on
the four variables than variety. That is, each variable changes significantly
with a different sowdate, but not significantly with a different variety.*

Problem 2b-2d

```
# b)
# Two-way (S, V) MANOVA model:  $y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ijk}$ 
#  $y_{ijk}$ :  $k$ th random observation corresponding to  $i$ th level of S,  $j$ th of V,
#  $\mu$ : hypothetical constant mean vector,  $\alpha_i$ : S-factor effect  $i$ th level,
#  $\beta_j$ : V-factor effect  $j$ th level,  $\gamma_{ij}$  = interaction  $i$ th of S and  $j$ th of V,
#  $\epsilon_{ijk}$ : error for each term  $\sim N(0, \text{constant dispersion})$ 
#  $H_0$ :  $\mu_1 = \mu_2 = \mu_3$  or  $\tau_1 = \tau_2 = \tau_3 = 0$ 
# vs.  $H_1$ : At least one inequality
```

```
# c)
bean_manova = manova(cbind(early,sla,total,avg)~sowdate*variety, data=bean_data)
summary(bean_manova)
```

```
##              Df  Pillai approx F num Df den Df Pr(>F)
## sowdate      1 0.96616   378.32     4    53 <2e-16 ***
## variety      1 0.09158     1.34     4    53 0.2689
## sowdate:variety 1 0.05971     0.84     4    53 0.5052
## Residuals    56
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Based on the results, it can be concluded that the sowdate is significant in
# affecting differences in at least one of the explanatory variables, whereas
# variety and its interaction with sowdate are NOT significant in affecting any
# differences in the four explanatory variables between the groups
```

```
# d)
# The results are certainly consistent with the one-way ANOVA models for each
# variable in that sowdate was found to be very significant and variety not
# significant. The interaction between these two factors was also found to be
# insignificant. This makes sense because the individual results for each
# variable were very similar, implying there is no significant interaction
# between sowdate and variety.
```

Problem 3

```
spectral = read.csv('C:/Users/jacma/Downloads/Spectrophoto_HW4Q3Data.csv')
x1 = spectral[,1] # explanatory variables
x2 = spectral[,2]
species = spectral[,3] # groups
time = spectral[,4] # factors
replication = spectral[,5]

spec_manov = manova(cbind(x1,x2) ~ species*time, data=spectral)
summary(spec_manov, test="Wilks")
```

```
##              Df  Wilks approx F num Df den Df      Pr(>F)
```

```
## species      2 0.32499   10.935      4      58 1.125e-06 ***
## time         1 0.26621   39.968      2      29 4.632e-09 ***
## species:time 2 0.42843    7.653      4      58 5.063e-05 ***
## Residuals    30
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Both time and species seem to be significant factors affecting reflectance at the two wavelengths, in addition to the interaction between species and time at alpha = 5%. This is because the associated p-values for each factor are extremely small.

```
species_fit = lm(cbind(x1, x2) ~ species*time, data=spectral)
group_effect = manova(species_fit, type="III")
```

```
## Warning: In lm.fit(x, y, offset = offset, singular.ok = singular.ok, ...) :
## extra argument 'type' will be disregarded
```

```
summary(group_effect, multivariate=TRUE)
```

```
##              Df  Pillai approx F num Df den Df      Pr(>F)
## species      2 0.69464    7.982      4      60 3.152e-05 ***
## time         1 0.73379   39.968      2      29 4.632e-09 ***
## species:time 2 0.57494    6.052      4      60 0.0003694 ***
## Residuals    30
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The results of the regression model support the conclusion from above, where each of the factors and their interaction are found to be significant at alpha = 0.05. Further, the test used to determine the eigenvalue-based test statistic does not alter any of the results, as all four are significant for each factor. Of the three factors, the significance seems to be in the following descending order based on the p-values calculated. Time, species-time interaction, then species. In other words, it could be concluded that the time at which measurements were taken is the most significant factor in causing differences between the two wavelengths.